

From Infinite Solutions to the One You Need

A ground-up companion to the Lecture 2 slides — initial-value problems, existence & uniqueness, and reading a solution's shape without solving anything.

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Quick Recall: Solutions Come in Families

TL;DR

Every DE you solve hands back a whole family of curves — one arbitrary constant, endless possibilities.

Solve a first-order equation like $\frac{dy}{dx} = 2x$ and integration gives $y = x^2 + C$ — not one curve, but an entire stack of parabolas, one for every value of C . That's normal: differentiating erases constants, so undoing a derivative always leaves one behind.

The question this lecture answers: out of that whole family, how do you pin down the *one* curve that actually matches your situation?

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Initial-Value Problems: Pinning Down One Curve

TL;DR

An IVP is a DE plus a snapshot value — that snapshot is what selects exactly one member of the family.

An **initial-value problem (IVP)** pairs a differential equation with one or more conditions that pin down the value of the unknown function (and maybe its derivatives) at a single point. On some interval containing x_0 :

$$\begin{cases} \text{solve:} & \frac{d^n y}{dx^n} = f(x, y, y', \dots, y^{(n-1)}) \\ \text{subject to:} & y(x_0) = y_0, y'(x_0) = y_1, \dots, y^{(n-1)}(x_0) = y_{n-1} \end{cases}$$

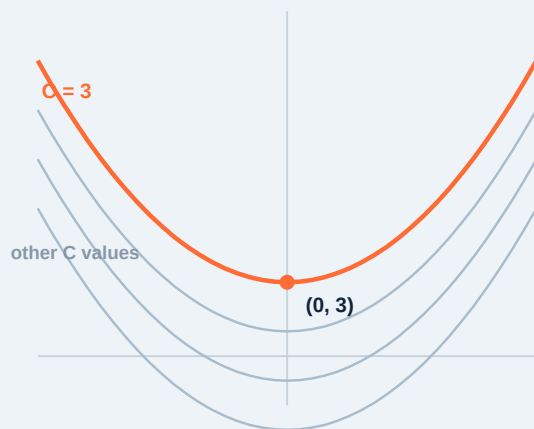
Here $y_0, y_1, \dots, y_{n-1}$ are just numbers you're handed — real constants specified in the problem. Together with the point x_0 , they're called **initial conditions (IC)**. An n th-order equation needs n of them, matching the n arbitrary constants that show up when you solve it.

LET'S WORK ONE END TO END

Take the family from Section 0: $\frac{dy}{dx} = 2x$, general solution $y = x^2 + C$. Now attach a condition: $y(0) = 3$.

Apply it: $3 = 0^2 + C \Rightarrow C = 3$. That's the whole trick — plug the known point into the general solution and solve for the leftover constant.

Particular solution: $y = x^2 + 3$ — exactly one parabola out of the infinite stack, the only one passing through $(0, 3)$.



The family $y = x^2 + C$ — the initial condition $y(0) = 3$ singles out the orange curve.

A SECOND-ORDER CASE, WORKED OUT

Second-order equations need *two* conditions, since solving one introduces two constants. Take $x'' + 16x = 0$, subject to $x(\pi/2) = -2$ and $x'(\pi/2) = 1$.

General solution (a two-parameter family): $x = c_1 \cos(4t) + c_2 \sin(4t)$.

Apply $x(\pi/2) = -2$: $-2 = c_1 \cos(2\pi) + c_2 \sin(2\pi) \Rightarrow c_1 = -2$

Now $x = -2 \cos(4t) + c_2 \sin(4t)$ and its derivative $x' = 8 \sin(4t) + 4c_2 \cos(4t)$.

Apply $x'(\pi/2) = 1$: $1 = 8 \sin(2\pi) + 4c_2 \cos(2\pi) \Rightarrow c_2 = \frac{1}{4}$

Particular solution: $x = -2 \cos(4t) + \frac{1}{4} \sin(4t)$

WHY BOTHER?

Order tells you how many conditions you'll need before you're stuck with a single, fully-determined curve. First-order $\rightarrow$ one condition. Second-order $\rightarrow$ two. No exceptions.

2 Does a Solution Always Exist? (And Is It the Only One?)

TL;DR

Not every IVP is guaranteed a solution — and even when one exists, it might not be the only one.

Two fundamental questions come up for every IVP, using $\frac{dy}{dx} = f(x, y)$ as the running example:

- **Existence** — does a solution curve even pass through the point (x_0, y_0) ?
- **Uniqueness** — if one does, can we be sure it's the *only* one?

A CAUTIONARY CASE

Take $\frac{dy}{dx} = xy^{1/2}$, subject to $y(0) = 0$.

Two different functions both check out: $y = \frac{1}{16}x^4$ **and** $y = 0$. Differentiate either one, substitute back, and both satisfy the equation — and both pass through $(0, 0)$.

So this IVP has a solution — it just isn't unique. Something about this equation makes multiple curves converge at the same starting point.

THE FIX: A THEOREM THAT GUARANTEES UNIQUENESS

Theorem. Let R be a rectangular region in the xy -plane, $a \leq x \leq b$, $c \leq y \leq d$, containing (x_0, y_0) in its interior. If $f(x, y)$ and $\partial f / \partial y$ are both continuous on R , then there exists some interval $I_0 : (x_0 - h, x_0 + h)$, $h > 0$, on which a solution exists — and it's the *only* one.

In plain terms: check that f and its partial derivative with respect to y are both "nice" (continuous) near your starting point. If they are, you're guaranteed exactly one solution curve there — no surprises like the case above.

3 Solution Curves Without a Solution: Direction Fields

TL;DR

You can sketch a DE's entire behavior with tiny slope arrows — no integration required.

Here's the qualitative shortcut: $\frac{dy}{dx} = f(x, y)$ tells you the *slope* a solution curve must have at any point (x, y) — like a compass needle sitting at every location on the plane, pointing which way the curve has to lean as it passes through.

COMPUTING ONE SLOPE

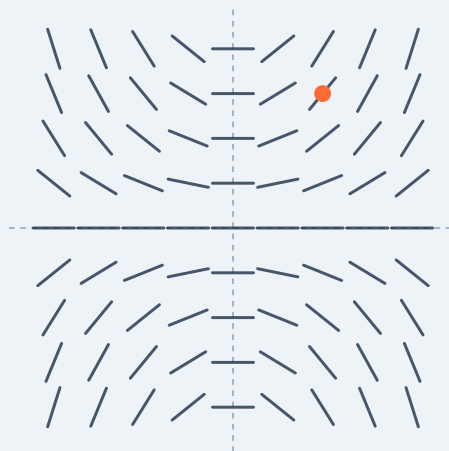
Take $\frac{dy}{dx} = 0.2xy$. At the point $(x, y) = (2, 3)$:

$$\frac{dy}{dx} = 0.2(2)(3) = 1.2$$

Any solution curve that happens to pass through $(2, 3)$ *must* be climbing with slope 1.2 right there — that's a hard constraint, whether or not you ever solve the equation.

PUTTING IT TOGETHER: THE DIRECTION FIELD

Repeat that slope calculation at a grid of points and draw a short line segment (a "lineal element") at each one. The whole collection is called a **direction field** or **slope field** — and any actual solution curve has to flow along the arrows, like a leaf carried by a current.



Direction field for $dy/dx = 0.2xy$ — the orange dot marks $(2, 3)$, slope 1.2.

Zoom out far enough on this particular field and the curves it traces turn out to be $y = ce^{0.1x^2}$ — but notice you never needed that formula to sketch the shape. The field alone tells you: curves flatten near $y = 0$ and steepen as $|x|$ and $|y|$ grow, symmetric across both axes.

LESSON

Direction fields matter most exactly when a DE *can't* be solved by hand. The picture still tells you where solutions rise, fall, or level off — for free.

4

Autonomous Equations: When Only y Matters

TL;DR

If x doesn't appear on the right-hand side — $\frac{dy}{dx} = f(y)$ — you can read off long-term behavior from algebra alone.

An ODE where the independent variable never shows up explicitly is called **autonomous**. Compare $\frac{dy}{dx} = 0.2xy$ (has x in it — not autonomous) against $\frac{dy}{dx} = 1 + y^2$ (only y appears — autonomous).

A real number c is a **critical point** (or equilibrium point) of $\frac{dy}{dx} = f(y)$ if $f(c) = 0$. At a critical point the slope is exactly zero everywhere along $y = c$, so $y(x) = c$ is itself a valid constant — or **equilibrium** — solution.

LET'S WORK ONE END TO END

Take $\frac{dy}{dx} = y(1 - y)$. Setting $f(y) = y(1 - y) = 0$ gives two critical points: $y = 0$ and $y = 1$. Those are two flat, constant solutions sitting right there in the family.

What about everywhere else? Check the sign of $f(y)$ on each interval between and around the critical points:

INTERVAL	SIGN OF $F(Y)$	$Y'(X)$	BEHAVIOR
$(-\infty, 0)$	–	< 0	decreasing ↓
$(0, 1)$	+	> 0	increasing ↑
$(1, \infty)$	–	< 0	decreasing ↓

Draw this as a **phase portrait**: a single vertical axis with the critical points marked, and arrows showing which way nearby solutions drift.

Solutions above 1 drift down toward it; solutions between 0 and 1 climb up toward it; solutions below 0 keep falling away from it.



5

Classifying Critical Points

TL;DR

The arrows around a critical point tell you whether nearby solutions get pulled in, pushed away, or stuck halfway.

Look at the two arrows immediately above and below a critical point c and read off which of four patterns you've got:



(a) **Attractor**
stable



(b) **Repeller**
unstable



(c) **Semi-stable**
mixed



(d) **Semi-stable**
mixed

PATTERN	CLASSIFICATION	MEANING
$\downarrow \bullet \uparrow$	Asymptotically stable — attractor	Solutions move toward c from both sides
$\uparrow \bullet \downarrow$	Unstable — repeller	Solutions move away from c on both sides
$\uparrow \bullet \uparrow$	Semi-stable	Toward c on one side, away on the other
$\downarrow \bullet \downarrow$	Semi-stable	Toward c on one side, away on the other

BACK TO $Y' = Y(1-Y)$

At $y = 1$: arrows climb toward it from below and fall toward it from above — a full **attractor**. Every solution starting anywhere above 0 eventually settles at $y = 1$. At $y = 0$: arrows fall away below it and climb away above it — a **repeller**. Nothing stays there for long.

WHY BOTHER?

This is prediction without solving. A population model with an attractor at its carrying capacity tells you the long-run outcome instantly — no formula required, just the sign pattern of $f(y)$.



The Big Picture

An initial-value problem is a differential equation plus enough starting data to select exactly one curve from an otherwise infinite family — one condition per order of the equation. That selection isn't automatically guaranteed: existence and uniqueness depend on $f(x, y)$ and $\partial f / \partial y$ behaving continuously near the starting point, and when they don't, multiple solutions can share the same starting point.

When solving outright isn't practical, direction fields let you read off a solution's shape directly from the equation — and for autonomous equations, critical points and phase portraits go one step further, revealing long-term behavior (attractor, repeller, or semi-stable) from nothing more than the sign of $f(y)$.

Based on: ENGR 213, Lecture 2 (Sept 11, 2026), Prof. A. Haghighat M. — reference text: Zill, Advanced Engineering Mathematics (7th ed.), Sections 1.2, 2.1. Worked examples marked "let's work one end to end" are original, built to illustrate the same ideas from the lecture.